

Wudneh Tilahun Mengist

Mathematics Lecturer & Researcher

Debre Tabor University · Department of Mathematics · Debre Tabor, Amhara, Ethiopia

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RESEARCH INTERESTS

Numerical analysis · Spline collocation methods · Singularly perturbed boundary value problems · Burgers' equation · Computational mathematics · Blended learning and Open edX instructional design

EDUCATION

Master of Science (MSc) in Mathematics 2008 — 2010

Bahir Dar University

Specialization: Numerical Analysis

Bachelor of Science (BSc) in Mathematics 2003 — 2007

Arba Minch University

Specialization: Applied Mathematics

ACADEMIC APPOINTMENTS

Lecturer in Mathematics 2011 — Present

Debre Tabor University

Undergraduate teaching across Numerical Analysis, Calculus, Linear Algebra, Differential Equations, and Number Theory. Active in faculty development and digital learning initiatives.

Layout Editor 2023 — Present

Journal of Interdisciplinary Science & Technology (JIST)

Typesetting, design, and editorial standards for the inaugural volume. LaTeX template design and brand identity.

Master / Field Trainer 2021 — 2026

National Election Board of Ethiopia

Operational training and leadership across regional training cycles.

PUBLICATIONS

Peer-Reviewed Journal Article

Mengist, W. T., et al. (2019). An exploration of quintic Hermite splines to solve Burgers' equation. *Arabian Journal of Mathematics*, 9(2), 351–367. <https://doi.org/10.1007/s40065-019-0247-y>

Metrics: 1,212 accesses · 13 citations · Published in Springer Nature

Manuscript Under Review

Mengist, W. T. (2025, under review). An adaptive B-spline collocation framework for singularly perturbed boundary value problems with boundary layers. *Journal of Interdisciplinary Science and Technology (JIST)*. <https://www.dtujust.com/>

Technical Reports & Frameworks

Mengist, W. T. (2024). Enhancing research and manuscript writing capabilities for secondary school and college mathematics teachers [Regional training framework].

TEACHING EXPERIENCE

Undergraduate courses taught at Debre Tabor University (2011 — Present):

MATH 2011 Numerical Analysis

MATH 1011 Calculus I — Differential
MATH 1012 Calculus II — Integral
MATH 2031 Number Theory
MATH 2041 Linear Algebra
MATH 3051 Differential Equations

DIGITAL LEARNING & OUTREACH

Creator & Educator

2020 — Present

The Hybrid Math Hub (YouTube)

Conceptualize, script, and produce video tutorials on numerical analysis, blended learning, multimedia design principles, and UDL-based course redesign.

IDL T Trainer

2024

Interactive Digital Learning & Teaching, Debre Tabor University

Facilitated courses and mentored faculty on Open edX integration and modern digital learning frameworks.

CERTIFICATIONS & PROFESSIONAL TRAINING

Financial Services & Capital Markets — *Ethiopian Securities Exchange (ESX) Digital Academy*

English Language Improvement Program (ELIP) — *DTU Academic Development*

TECHNICAL COMPETENCIES

Mathematical Computing: MATLAB, Python, R & SPSS, Wolfram Mathematica

Typesetting & Design: LaTeX (Overleaf), Graphic Design, Document Layout

E-Learning & Pedagogy: Open edX, Instructional Design, Video Production

Numerical Methods: Spline Collocation, Burgers' Equation, Singular Perturbation, Finite Differences

PROFESSIONAL AFFILIATIONS

ORCID — Open Researcher and Contributor ID (0000-0002-4335-3741)

JIST — Journal of Interdisciplinary Science and Technology (Editorial Board)

LANGUAGES

Amharic (Native) · **English** (Professional working proficiency)

REFERENCES

Available upon request. Please contact via email at wudneh.tilahun@dbtu.edu.et or via the contact form on the portfolio website.